

Initial reproduction report and feasibility audit

Audit date: 7 October 2026. This report precedes final analysis and distinguishes published methods, official code and our extension.

Selection decision

Rodrigues et al., **A Data Rate Monitoring Approach for Cyberattack Detection in Digital Twin Communication**, Sensors 25(24), 7476, published 9 December 2025, DOI [10.3390/s25247476](https://doi.org/10.3390/s25247476), is the selected reference baseline. Its publication is independently verified by [PubMed](#). It is a recent peer-reviewed journal article, not claimed to be a top-tier systems venue. The [official repository](#) is linked in the article's data-availability statement and identifies the same article in its README. It fits the existing topic through MQTT-mediated physical/virtual telemetry and communication resilience. Its security focus differs from our recovery/freshness contribution; we retain that distinction.

Published-method summary

The reference monitors communication rates using a mean-plus-three-standard-deviations threshold. Its Docker topology connects a synthetic device, MQTT broker, twin, monitor and controlled attack components. Reported observables are PPS/BPS; the healthy recording is described as 60 minutes. The paper specifies Ubuntu 24.04.3, Docker 28.4.0, Compose 2.39.4, Python 3.12, Paho 1.6.1, Requests 2.32.3, OPC-UA 0.98.13 and aiocoap 0.4.3. It evaluates DoS, active MiTM and intrusion; Table 2 provides scenario statistics. A precise CPU/RAM specification, original packet captures and repeat-run seeds were not located. The publicly readable [article record](#) and the Europe PMC full-text XML were checked. The containerized twin is a deployable service; the published evaluation is not evidence of deployment on a public cloud.

Official-artifact audit (findings from code inspection)

Pinned commit: 43346f60a9f41392611242ff3cb7589d7a758aa7.

Check	Finding	Consequence
Source provenance	Article data-availability link matches repository/README	Official research artifact verified
License	No project-level LICENSE found; only bundled dependency licenses	Fetch privately; do not redistribute/relicense author code
Data	Synthetic device generator; no separate author CSV/PCAP found outside committed venv	Generate our own traces; Table 2 cannot be exactly recomputed
Dependencies	Dockerfiles use Python 3.10-slim and mostly unpinned pip installs	Different from paper's Python 3.12; native adapter pins packages
Paho	Source uses CallbackAPIVersion, unavailable in paper's specified 1.6.1	Use 2.1.0; document discrepancy
Detection rule	stats_monitor compares rates to calibration maxima	Preserve code behavior; independent paper-rule implementation alongside it
Capture boundary	MQTT callbacks increment message counts and payload bytes	These are application messages/s and payload bytes/s, not all wire packets/bytes

Check	Finding	Consequence
Calibration	Device sleeps $15 + 3 \times (15+2) = 66$ seconds, disabling sensors in turn	Different from healthy 60-minute recording
Completion signal	Native attempt did not receive final calibration message before publisher disconnect	Failed log retained; adapter resends only this message with network loop/QoS1
Native log encoding	Unicode arrows fail with default Windows file encoding	Adapter writes UTF-8 logs
State storage	Receiver prints payloads; timestamp/sequence ordering is absent	Extension explicitly adds timestamped current-state semantics

Candidate comparison

Candidate	Advantages	Reason not selected as executable primary
An et al., ICT Express 2025	Direct MQTT/UAV digital-twin topic and recent journal	Official code/dataset not located; publisher full-text request returned 403
Cakir et al., IEEE CAMAD 2023	Direct synchronization study and alignment metric	Older; no official simulation code found in checked sources; ns-3 topology not equivalent to MQTT buffering
Dizdarevic et al., 2023 preprint	Author-linked MQTT broker testbed , useful impairment methodology	Peer-reviewed venue not verified; KVM/Nebula/JMeter/hardware requirements; README inherited BenchFaaS content
ComBench, VALUETOOLS 2021	Official code , configurable workloads	Older and protocol benchmarking rather than a recent twin application
Neagu et al., Future Internet 2026	Recent operational-twin resilience	Node failover rather than MQTT current-state replay; official executable repository not located in checked sources

These searches establish feasibility within inspected sources, not proof that no unpublished code exists. Newer IEEE freshness literature informs the contribution even though its optimization algorithms are not the executable baseline.

Reproduction scope

Run original sensor, MQTT receiver, MQTT monitor and rate-monitor functions with a native adapter, retaining original bodies and calibration timing. Replace the broker with loopback AMQTT and omit unrelated multiprotocol servers. Exercise a benign high-rate sensor burst. Record code hashes, observer events, monitor logs and adaptation details. Implement the paper's statistical rule independently and compare it with the source's maximum rule.

Do not claim: exact Table 2 reproduction, attack-detection accuracy, public-cloud performance, real packet-loss experiments, physical PLC tests, Raspberry Pi energy measurements or all upstream protocols. Docker engine was unavailable initially and remained unresponsive after a launch attempt. Linux tc/netem is not available for this Windows native test. No measured value is copied from the paper as a result of our runs.

Contribution hypothesis

A rate-only monitor and FIFO replay can treat many old updates as successful communication while the current state remains stale. Test a latest-state outbox with fair service and a monotonic receiver against FIFO, epoch-only staging and expiry. Retain all generated updates in a separate local audit log; count intentional live-update suppression transparently. This is a scoped systems evaluation built on known freshness ideas, not a claim to have invented Age of Information or last-generated-first-served scheduling.